

P8139 Homework 3

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Problem 1

Read in the math/chem/bio scores data for PCA practice and answer the following questions.

```
mydata = read.table(file = "PCA_data.csv", header = TRUE, row.names = 1, sep = ",")
mydata.pca = prcomp(mydata, retx = TRUE, center = TRUE, scale = TRUE)

sd = mydata.pca$sdev
loadings = mydata.pca$rotation
rownames(loadings) = colnames(mydata)
scores = mydata.pca$x
```

- (a) Calculate PCA scores using loadings and original math/chem/bio scores and compare to output PCA scores from the R package prcomp.

```
# Standardize the original data
Z = scale(mydata, center = mydata.pca$center, scale = mydata.pca$scale)

# Compute PCA scores manually
manual_scores = Z %*% loadings
manual_scores
```

##		PC1	PC2	PC3
## 1		-1.6579369	0.571682476	0.27659132
## 2		-2.5759191	0.171154076	-0.39465359
## 3		-1.8500938	0.561523155	-0.32895929
## 4		-2.0029634	-0.279232617	0.20909576
## 5		-1.7019579	-0.553776982	0.53149553
## 6		-1.0237890	0.746472175	0.42698709
## 7		-0.9343334	0.599025432	0.24836634
## 8		-2.0297283	-0.031871212	-0.13372583
## 9		-2.2666815	0.246059593	-0.25427541
## 10		-2.6579342	-0.005870397	-0.16437182
## 11		1.3079701	0.359969524	0.41720262
## 12		0.9980941	1.269880082	-0.28446005
## 13		1.3110519	1.182549991	-0.11825137
## 14		1.6046169	1.232476842	0.15248740
## 15		1.2290368	1.005525517	0.11203039
## 16		0.8027869	0.651507049	0.04113399
## 17		0.8884538	0.880662669	0.23237339

```
## 18  1.5032773  0.978342557  0.09160858
## 19  1.1241139  0.484893331 -0.76606024
## 20  0.7141229  1.472874891 -0.01392787
## 21  1.1276809 -1.336079451  0.05473737
## 22  0.3206321 -1.229551905 -0.01435777
## 23  0.9474764 -1.159024498 -0.70700433
## 24  0.5478545 -1.331670861  0.35630217
## 25  1.3013734 -0.943475598 -0.26024587
## 26  0.6022446 -1.366839327  0.41257208
## 27  1.0307847 -0.864161491  0.18169711
## 28  0.4413646 -1.112674562 -0.05800904
## 29  0.4666995 -1.049140991 -0.04278934
## 30  0.4317024 -1.151229468 -0.20358933
```

```
cat("\nMax absolute difference:", max(abs(manual_scores - scores)), "\n")
```

```
##
## Max absolute difference: 0
```

The manually computed scores are identical to the `prcomp` output, confirming that PCA scores = standardized data $\times$ loadings.

(b) Calculate percent variance explained of each component.

The variance explained by each PC equals its eigenvalue (square of `sdev`), and the percent is the eigenvalue divided by total variance.

```
eigenvalues = sd^2
pct_var = eigenvalues / sum(eigenvalues) * 100

var_table = data.frame(
  Component = paste0("PC", 1:length(sd)),
  Eigenvalue = round(eigenvalues, 4),
  Pct_Variance = round(pct_var, 2),
  Cumulative_Pct = round(cumsum(pct_var), 2)
)
print(var_table)
```

```
##   Component Eigenvalue Pct_Variance Cumulative_Pct
## 1      PC1      2.0078         66.93           66.93
## 2      PC2      0.8913         29.71           96.64
## 3      PC3      0.1008          3.36          100.00
```

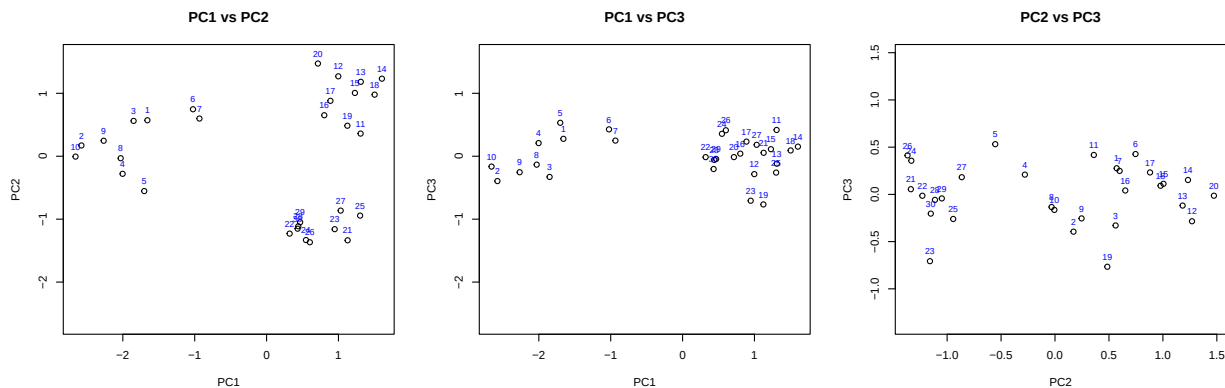
(c) Plot i) PC1 vs PC2, ii) PC1 vs PC3, and iii) PC2 vs PC3.

```
par(mfrow = c(1, 3))

# i) PC1 vs PC2
plot(scores[, 1], scores[, 2], xlab = "PC1", ylab = "PC2", main = "PC1 vs PC2",
      xlim = range(scores[, 1:2]), ylim = range(scores[, 1:2]))
text(scores[, 1], scores[, 2], rownames(scores), col = "blue", cex = 0.7, pos = 3)
```

```
# ii) PC1 vs PC3
plot(scores[, 1], scores[, 3], xlab = "PC1", ylab = "PC3", main = "PC1 vs PC3",
     xlim = range(scores[, c(1,3)]), ylim = range(scores[, c(1,3)]))
text(scores[, 1], scores[, 3], rownames(scores), col = "blue", cex = 0.7, pos = 3)

# iii) PC2 vs PC3
plot(scores[, 2], scores[, 3], xlab = "PC2", ylab = "PC3", main = "PC2 vs PC3",
     xlim = range(scores[, 2:3]), ylim = range(scores[, 2:3]))
text(scores[, 2], scores[, 3], rownames(scores), col = "blue", cex = 0.7, pos = 3)
```



The PC1 vs PC2 plot clearly separates three clusters of students: those strong in math only (students 1–10), those strong in bio/chem (students 11–20), and those strong in all subjects (students 21–30). PC1 captures overall academic ability, while PC2 distinguishes math-oriented vs. bio/chem-oriented students.

Problem 2

Spielman et al. (1993) introduced the **Transmission Disequilibrium Test (TDT)**, a family-based method for testing linkage between a marker locus and a disease locus. The key innovation of the TDT is that it provides a valid test for *linkage* that is not confounded by population stratification, unlike conventional population-based association tests.

Background. A population association between IDDM and “class 1” alleles of the 5’FP on chromosome 11p had been consistently observed, but earlier ASP-based linkage studies failed to find evidence for linkage. Since association alone cannot establish linkage (it may reflect population stratification), the authors developed a family-based test that directly tests for linkage.

The TDT Method. The TDT focuses on **heterozygous parents** (M_1M_2) who have at least one affected offspring. It compares the number of times the associated marker allele (M_1) is transmitted (b) versus not transmitted (c) to affected offspring. The test statistic is McNemar’s χ^2 :

$$\chi_{td}^2 = \frac{(b - c)^2}{b + c}$$

which follows a χ^2 distribution with 1 df under the null hypothesis of no linkage ($\theta = 1/2$). The authors demonstrated that the conventional χ^2 comparing transmitted vs. nontransmitted allele frequencies (AFBAC method) is only valid for testing no association ($\delta = 0$), not for testing no linkage. The TDT is valid because, as the paper states, “only the data from heterozygous M_1M_2 parents should be used in the test,” and the statistic depends on both linkage and linkage disequilibrium ($\delta > 0$). This design is inherently robust to population stratification because each heterozygous parent serves as their own control: under

H_0 , the probability of transmitting either allele is exactly 1/2 regardless of population allele frequencies, so population-level differences in allele frequency do not enter the test.

Application to IDDM. Applied to GAW5 data, the TDT yielded $\chi^2_{td} = 8.26$ ($p = .004$), providing strong evidence for linkage, whereas haplotype-sharing analysis of the same data found none ($\chi^2_{hs} = 0.20$). Note that the TDT requires LD to detect linkage — “*even very close linkage will not be detected by the TDT if $\delta = 0$* ” — so it is best used when a population association has already been found.

Relevant Quotes. Several passages clarify concepts from class:

- On population stratification: “*Population association between a disease and a genetic marker can arise as an artifact of population structure, even in the absence of linkage.*”
- On LD: The paper parameterizes LD as $\delta = \text{freq}(M_1D_1) - mp$, and Table 3 shows the TDT depends on both δ and θ , formalizing LD as a population-level concept distinct from within-family linkage.
- On samples for linkage studies: “*The distribution of 5'FP alleles in 33 affected sib pairs (ASPs) with IDDM... failed entirely to provide evidence for linkage.*” This contrasts ASP studies (multiplex families) with the TDT (simplex families with heterozygous parents).